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VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

K. J. Somaiya College of Engineering, Mumbai-77

(A Constituent College of Somaiya Vidyavihar University)



Department of Electronics Engineering

Course Name:	Object Oriented Programming	Semester:	III
Date of Performance:	___ / ___ / ____	Batch No:	A4
Faculty Name:	Dr. Tejashree Phatak	Roll No:	15015024082
Faculty Sign & Date:		Grade/Marks:	___/15

Experiment No: 3

Title: Concept of static method and recursive function

Aim and Objective of the Experiment:

Learn the concept of static method and recursive function in Java

COs to be achieved:

CO1: Understand concepts of Object Oriented Programming and basic characteristics of Java.

Tools used:

Theory:



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1. (About static method)

static methods are stored in memory only once at the beginning of the program and is executed when called, a single instance can be called without creating object of the class.

2.

Code:



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1. Write a recursive function 'GCD' to find the GCD of the given two numbers. Use this in main to find the GCD and LCM two given numbers.

- Variation 1: Implementation with One class only
- Variation 2: Accessibility with static and non-static methods within class and outside

class.

```
class GCDLCM {
```

```
Var 1 :
```

```
class GCDLCM {
```

```
// Recursive function for GCD
```

```
static int gcd(int a, int b) {
```

```
    if (a == 0)
```

```
        return b;
```

```
    return gcd(b % a, a);
```

```
}
```

```
// LCM using GCD
```

```
static int lcm(int a, int b) {
```

```
    return (a / gcd(a, b)) * b;
```

```
}
```

```
public static void main(String[] args) {
```

```
    if (args.length < 2) {
```

```
        System.out.println("Please provide two integers as command-line arguments.");
```

```
        return;
```

```
    }
```

```
    int num1 = Integer.parseInt(args[0]);
```

```
    int num2 = Integer.parseInt(args[1]);
```

```
    System.out.println("Numbers: " + num1 + " and " + num2);
```

```
    int g = gcd(num1, num2);
```

```
    int l = lcm(num1, num2);
```

```
    System.out.println("GCD = " + g);
```

```
    System.out.println("LCM = " + l);
```

```
    }
```

```
}
```



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Var 2 :

```
class MathOperations {

    public static int gcd(int a, int b) {
        if (a == 0)
            return b;
        return gcd(b % a, a);
    }

    public int lcm(int a, int b) {
        return (a / gcd(a, b)) * b;
    }
}

public class GCDLCM {
    public static void main(String[] args) {
        if (args.length < 2) {
            System.out.println("Please provide two integers as command-line arguments.");
            return;
        }

        int num1 = Integer.parseInt(args[0]);
        int num2 = Integer.parseInt(args[1]);

        int g = MathOperations.gcd(num1, num2);

        MathOperations obj = new MathOperations();
        int l = obj.lcm(num1, num2);

        System.out.println("Numbers: " + num1 + " and " + num2);
        System.out.println("GCD = " + g);
        System.out.println("LCM = " + l);
    }
}
```

Output:



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```
stella>javac GCDLCM.java
java GCDLCM 56 98
Numbers: 56 and 98
GCD = 14
LCM = 392
```

Post Lab Subjective/Objective type Questions:

1. Write a recursive static method for calculation of factorial of n number.

```
static int fact(int n) {
    if(n == 1) return 1;
    return n * fact(n-1);
}
```

2. What are command-line-argument and how are they used?

CL arguments are values passed to a Java program at runtime via the command line and are accessed through the String[] args parameter of the main method.

Conclusion:

Learned about static methods, how they are used and what the uses, their memory allocations, CL arguments- how they work, what are they, recursion in Java.

Signature of faculty in-charge with Date: